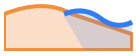
Which EO model is worth training first

The task-first plan after reading the two overlap papers: coastal first, forecasting second, burn search without training.

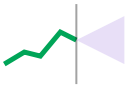
The plan in one line

Start with coastal change. The thing worth learning is how wrong each satellite shoreline is, checked against field surveys. Forecasting comes second. Burn-scar search needs no trained model.



1. Coastal: is the change real?

Train a small model on top of existing tools (CoastSat, CoastSeg) that predicts each image's shoreline error and says when to abstain.



2. Greenness forecast

Fine-tune Contextformer with radar input, but only if a cheap CPU test shows radar helps. Licence (CC BY-NC-SA) still to decide.

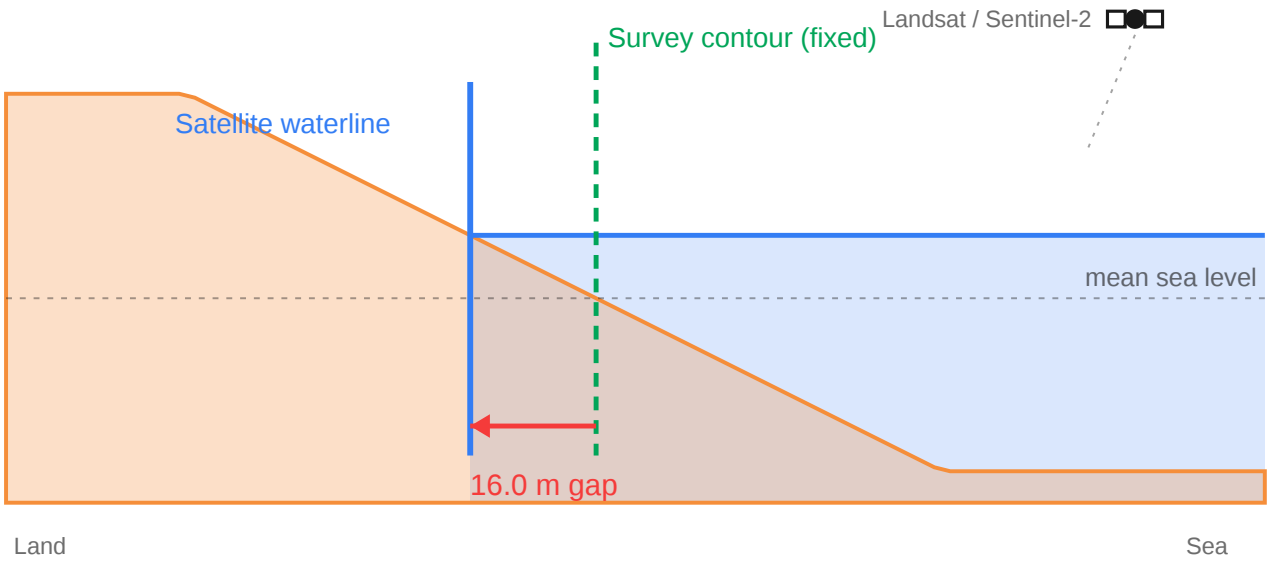


3. Burn-scar analogues

No training. Shape and severity measurements plus nearest-neighbour search. Labels from MTBS would make a trained model circular.

Why coastal: the waterline moves even when the beach doesn't

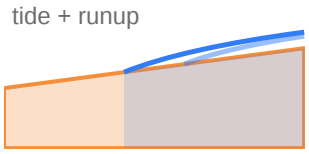
A satellite sees where the water is at that moment. A survey measures a fixed height on the beach. Move the tide and see how far apart they get.



Water level is +0.8 m, so the waterline sits **16.0 m landward** of the survey line. Compared with an image at mean sea level, this looks like **16.0 m of erosion**. The beach did not move.

The gap is water level divided by slope. Same physics as CoastSat's tide correction. The tide part can be corrected if you know the slope. What's left over (waves, a wrong slope, a shifted image, clouds) varies from image to image. Predicting that leftover for each image is the model. Numbers are illustrative, not measurements.

What else makes one image wrong



Tide and waves

Water sits higher or lower on the beach than the survey height.

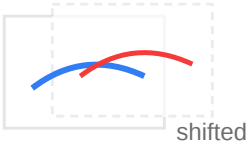


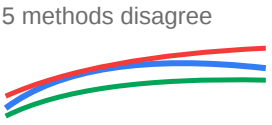
Image shifted

Images are not perfectly aligned to the ground. The RSMA paper found this part does not go away with contour tuning.



Clouds and shadows

Partly hidden coast or a bad water/sand threshold.

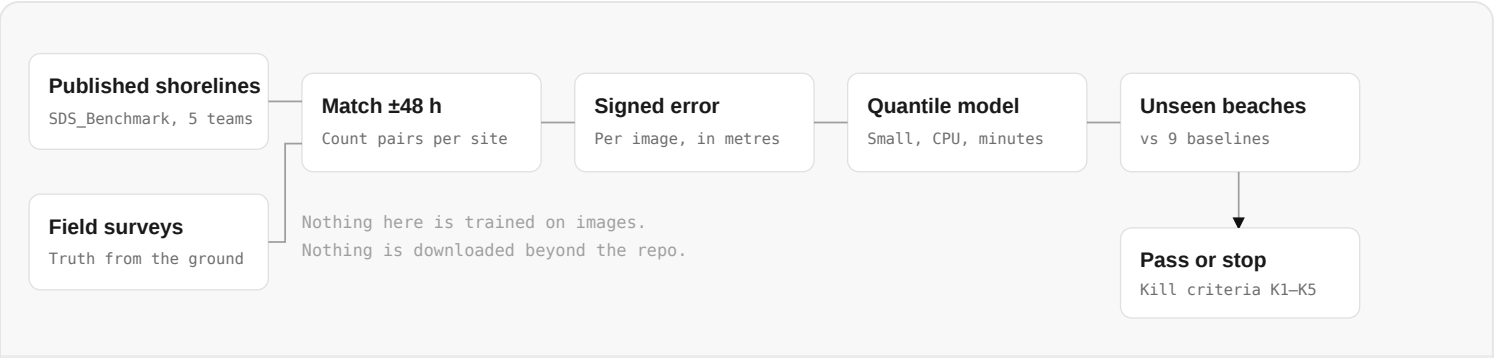


Method disagreement

The 5 SDS_Benchmark teams' lines spread apart on hard images. The spread is a useful warning sign.

The first test: Stage 0, on a laptop CPU, with no imagery

It reuses shorelines already published in SDS_Benchmark. Click a step for detail.



Select a step

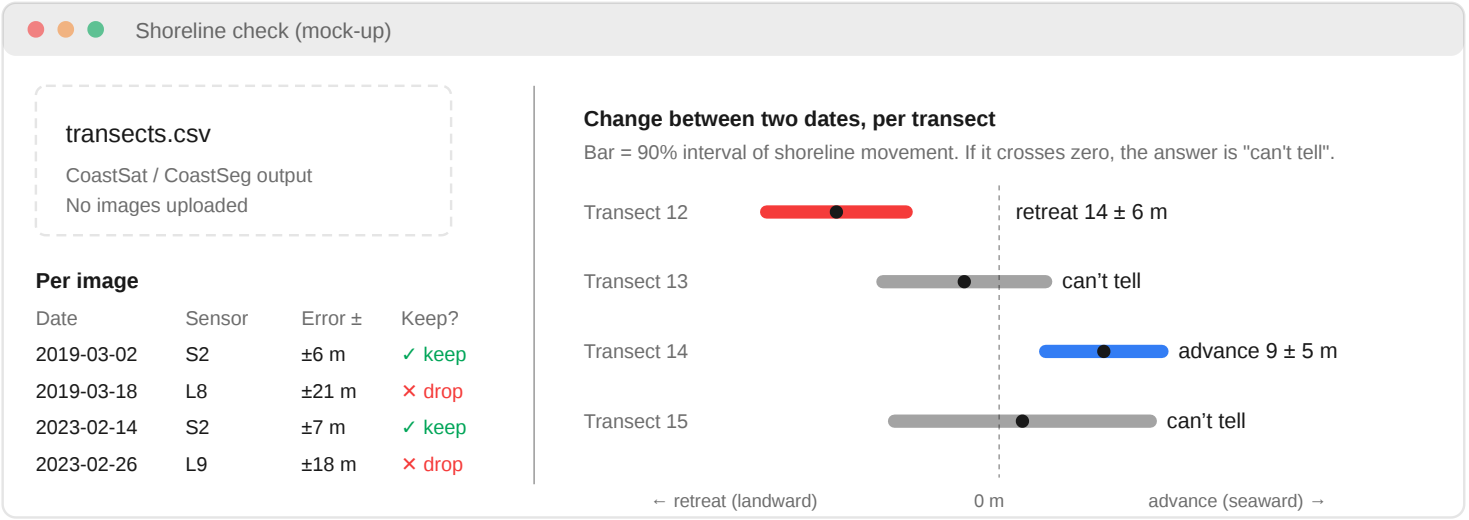
Each step lists what it uses and what would stop the project.

When to stop (coastal kill criteria)

Check	Stop or change course if…	What we'd do instead
K1	Less than 15% lower RMSE on kept images (70% kept), on 2 or more held-out sites	Ship a non-learned uncertainty tool
K2	90% intervals cover the truth less than 85% of the time	The tool may only abstain, not give intervals
K3	Wrong direction in more than 10% of claims, or fewer than 25% of date pairs get a claim	No "real change" verdicts. RSMA 2026 makes this more likely.
K4	Image embeddings add less than 5% over metadata alone	Drop the image encoder
K5	Fewer than 100 matches within ±48 h per test site	Widen the window and use the time gap as a feature, or stop

What the Hugging Face Space would show

A sketch, not a built app. Numbers are invented to show the layout.



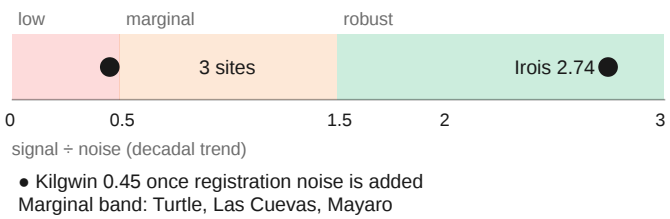
Each transect gets "retreat X ± Y m", "advance" or "can't tell". The Space runs on CPU. CoastSat/CoastSeg are GPL-3.0, so the Space code would have to be GPL-compatible. Tide grids (FES) are not redistributed.

What the two overlap papers changed

Neither kills its idea. Both narrow what we can claim as new.

Coastal: Aladejana et al. 2026

Regional Studies in Marine Science, doi:10.1016/j.rsma.2026.105374. Abstract read; full text is closed.



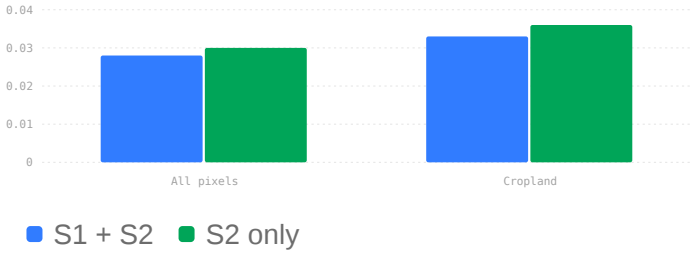
Only Irois is clearly above the noise. At Turtle, 82.7% of transects flipped direction depending on the rate used. The positions of the three marginal sites inside the band aren't given in the abstract.

- They already ask our question, but per beach and per decade, using a physical noise budget.
- From the abstract: no learned model, no beaches without surveys, no per-image error.

> **Change:** our claim becomes "predict each image's error, and transfer that to beaches with no surveys." Their noise budget is added as baseline 9.

Forecasting: Defonte et al. 2026

arXiv 2605.04239 (Thales, CNES). Full text read.



– Data table

Reflectance RMSE, lower is better (paper Table III, gap-filling and forecasting pooled)

Pixels	S1 + S2	S2 only
All pixels	0.028	0.030
Cropland	0.033	0.036

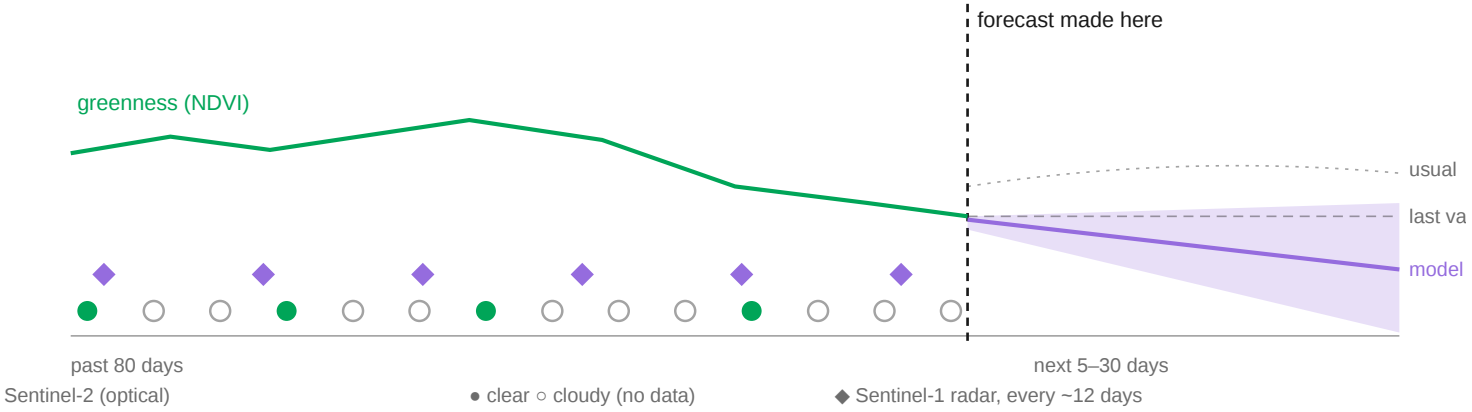
Radar helps, but only a little, on European farmland. The comparison mixes gap-filling with forecasting, and their forecasting results have no persistence or climatology baseline.

> **Change:** "radar helps" is no longer new in general. Ours must be forecast-only, per horizon (5–30 days), against persistence and climatology, on African rangeland. Their small gain also lowers our odds of passing K1.

The other two ideas, in pictures

Forecast greenness 5–30 days ahead, with radar filling cloud gaps

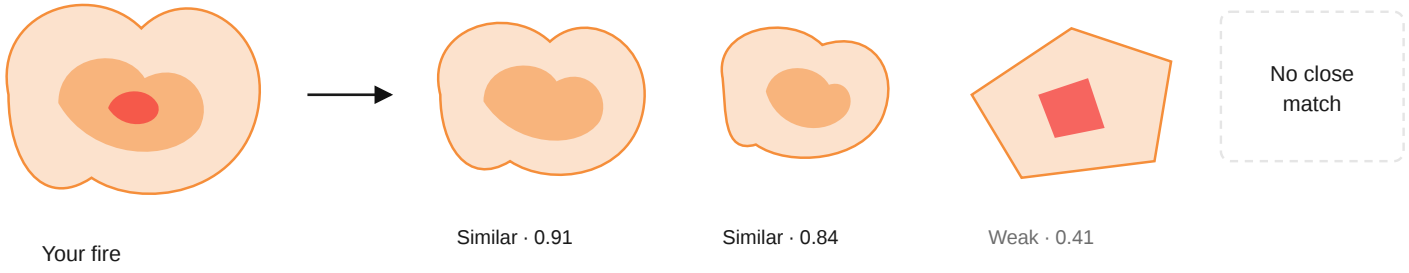
Clouds hide many optical images. Radar sees through clouds. Does radar history make the forecast better than just carrying the last value forward, or using the usual value for that time of year?



Shape is illustrative. Data would be EarthNet2023-Africa (CC BY-NC-SA). The model can't see future weather. The kill test: if radar doesn't lower NDVI error by at least 0.005 when optical images are sparse, drop it.

Find past fires that looked like this one, or say there are none

Give a burn footprint and severity map. Get back verified past US fires with a similar shape and severity, from MTBS. Answering "no close match" correctly matters as much as finding one.



Similarity scores are invented. The method is about 30 shape and severity measurements plus nearest neighbours, with a calibrated cut-off for "no match". First check: do two people grading pairs blind agree (kappa ≥ 0.4)? If not, stop.

Next decision

Waiting on you: approve Coastal Stage 0 (local CPU, clone SDS_Benchmark, no imagery, no GPU, no cluster). Forecasting Step A also needs a decision on whether a CC BY-NC-SA dataset is acceptable for a public Space.

Sources: TASK_FIRST_TRAINING_PLAN.md, OVERLAP_PAPERS_REVIEW.md and the three research-session reports. RSMA 2026 numbers come from its abstract; Defonte et al. numbers come from arXiv 2605.04239 Table III. Illustrations, mock-up values and similarity scores are invented to explain the ideas and are not results. Nothing has been trained, downloaded beyond small metadata, or deployed.